

Implementing USD: A Case Study in Incremental Adoption

Scott Milner
Brigham Young University
Computer Science
Provo, Utah, USA
scottdmilner@gmail.com

Stephanie Martins
University of Utah
Salt Lake City, Utah, USA
stephanie.martins@utah.edu

Conner Murray
University of Utah
Salt Lake City, Utah, USA
conner.murray@utah.edu

Zachary Wood
Brigham Young University
Computer Science
Provo, Utah, USA
zach.wood13@gmail.com

Craig Van Dyke
Brigham Young University
Computer Science
Provo, Utah, USA
craig@cs.byu.edu



Figure 1: Four years of BYU Animation productions with USD, 2022-2025

Abstract

Over the course of four productions, Brigham Young University (BYU) Animation has transitioned our animated short film pipeline to be fully centered on Pixar’s Universal Scene Description (USD). Each production set a goal of integrating USD in a way that built upon the previous productions’ work. Through these productions, we demonstrate an incremental USD adoption timeline suitable for small studio and educational contexts, resulting in a workflow that uses USD end to end. We present this process as a case study for how any small studio can implement USD into their pipeline.

CCS Concepts

• Applied computing → Collaborative learning; • Computing methodologies → Animation.

Keywords

USD, Animation, Pipeline, Education, Production

ACM Reference Format:

Scott Milner, Stephanie Martins, Conner Murray, Zachary Wood, and Craig Van Dyke. 2025. Implementing USD: A Case Study in Incremental Adoption. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educators Forum (SIGGRAPH Educators Forum '25)*, August 10–14, 2025, Vancouver, BC, Canada.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Educators Forum '25, Vancouver, BC, Canada
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1544-0/25/08
<https://doi.org/10.1145/3721242.3734008>

2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721242.3734008>

1 Introduction

For our past four productions, BYU Animation has worked to adopt USD into our animated short film pipeline. By adapting incrementally, we kept the workload manageable despite tight deadlines and small teams. Each of the four productions set a USD adoption goal that built upon previous productions’ work. Herein, we derive a USD adoption timeline suitable for both small studio and educational contexts. In *Cenote* (2022), we tackled saving assets and composing layouts with USD. Next, *The Witch’s Cat* (2023) built on top of *Cenote*’s work by lighting and rendering directly from USD. The following year, *Student Accomplice* (2024) used USD workflows to scale the scope of production, constructing multiple large environments and accommodating overlapping department timelines. Most recently, *Love & Dungeons* (2025) refined pipeline tools and workflows to optimize performance and use USD from end to end.

2 Assets and Layout

Cenote (2022) was our first production to utilize USD in its pipeline. The film had several environments that required multiple versions of the same asset, such as trees, rocks, and seaweed. We set a goal to use USD files for asset management, and take advantage of USD’s variant functionality.

We used USD as the main file type for geometry and materials. A Python script utilizing the USD API assembled the asset descriptions and organized variant sets. Layouts were built in Houdini Solaris. This allowed us to quickly switch between variants of each asset.

At the time, rendering from Solaris was not robust, so we created a tool to parse USD layout descriptions and unpack them for lighting and rendering in the Houdini Object context.

3 Rendering

The main goal of *The Witch's Cat* (2023) was to render our film with USD. The previous production struggled with long render times and disk space constraints from caching bulky render description files. Because *Cenote* used a traditional RenderMan for Houdini workflow, each frame's render description file needed to be re-cached each time it was sent for processing. *The Witch's Cat* had an even larger FX scope with three times the cloth and hair simulation, so rendering directly from USD was critical to meet our tight deadline on our small render farm.

USD enabled us to overcome these challenges by giving us deep control of where, when, and how various departments organized their assets and shot data. We made sure all animated data was properly cached before it was ingested by downstream departments. Because USD allowed us to use previously cached data directly in render, the time required to generate a viable render description to send to render dropped from hours of compute time to mere seconds. We rendered more in less time, giving our FX and lighting artists quicker iteration speed.

4 USD at Scale

Student Accomplish (2024) contained BYU's largest environments to date in terms of complexity and physical size. We built and iterated on large environments the size of city blocks that would have been prohibitively expensive without USD's instancing. USD's purpose system allowed us to swap between low level of detail (LOD) assets for scene layout and final LOD assets at render time. The USD Scene Graph structure made it easy to temporarily disable loading portions of heavy scenes when needed. The Solaris Component Builder was invaluable for managing assets and their variants in an artist-friendly abstraction. We constructed entire city's worth of unique buildings leveraging USD's variant and instancing capabilities.

Another benefit for the production was USD's parallelism. Given the scale of *Student Accomplish*, meeting deadlines required that every department work concurrently, even within the same shots. Saving department deliverables to USD layers allowed new updates to automatically flow downstream. For example, environments could be constructed with in-progress assets. USD's stable scene graph meant that lighting and effects artists could begin iteration early on shots without worrying about upstream changes causing lost work.

5 Optimizing a USD Pipeline

Love & Dungeons (2025) went into production with the goal of streamlining and optimizing the work that had been done previously towards USD adoption. This required a deeper understanding of the functionality and limitations of USD as a pipeline tool. For example, we changed the way we brought animation data into a scene to better match USD's structure paradigms. This kept our animation data organized and separate from our character data and allowed our animation exports to be lightweight, containing nothing but animated points. Other USD features allowed us to separate

Table 1: USD Deliverables in the Pipeline by Year

Deliverable (DCC)	2022	2023	2024	2025
Model (Maya)	OBJ	OBJ	USD	USD
Asset (Houdini)	USD	USD	USD	USD
Camera (Maya)	FBX	FBX	USD	USD
Animation (Maya)	ABC	ABC	ABC	USD
Layout (Houdini)	USD	USD	USD	USD
FX (Houdini)	BGEO	USD	USD	USD ¹
Render Desc. (Houdini)	RIB	USD	USD	USD

effects preroll into separate files, further reducing the animation data overhead for lighting and render.

We continued to use the Solaris Component Builder, adding auto-decimation and UV tile compression to our asset's preview LOD. This kept our environments performant, even in the Maya viewport with materials enabled. We found that begin/end blocks were visuals that Houdini artists are already familiar with, so we used them to help layout artists organize the scene graph. Clean and organized scene graphs saved the lighting team hours when polishing their shots. We also built tooling around the Solaris Hair Procedural, saving us hours of CFX caching and giving animators more control over the hair.

Altogether, *Love & Dungeons* achieved production on a similar scale to *Student Accomplish* while making significant performance improvements. This marks the successful implementation of a robust and efficient USD pipeline. Our plan for future productions includes finding better solutions for version management and variant tracking, possibly via USD Asset Resolvers.

6 Resources

We found the following resources to be useful while working on practical USD adoption with tools already provided by DCCs:

- SideFX's Houdini Solaris documentation
- Luca Scheller's *Usd Survival Guide*
- Remedy Entertainment's *Book of USD*
- Matt Estela (tokeru)'s *Usd Guide*

7 Conclusion

Over the course of four short film productions, we have determined that an incremental approach to USD adoption is accessible, beneficial, and possible even on tight deadlines and with a small team. USD is not merely a file format, but an approach to structuring a pipeline that promotes organization, facilitates departments working simultaneously, and provides significant potential for scaling up capacity across the production pipeline.

¹with Husk procedurals